

Experiment Report

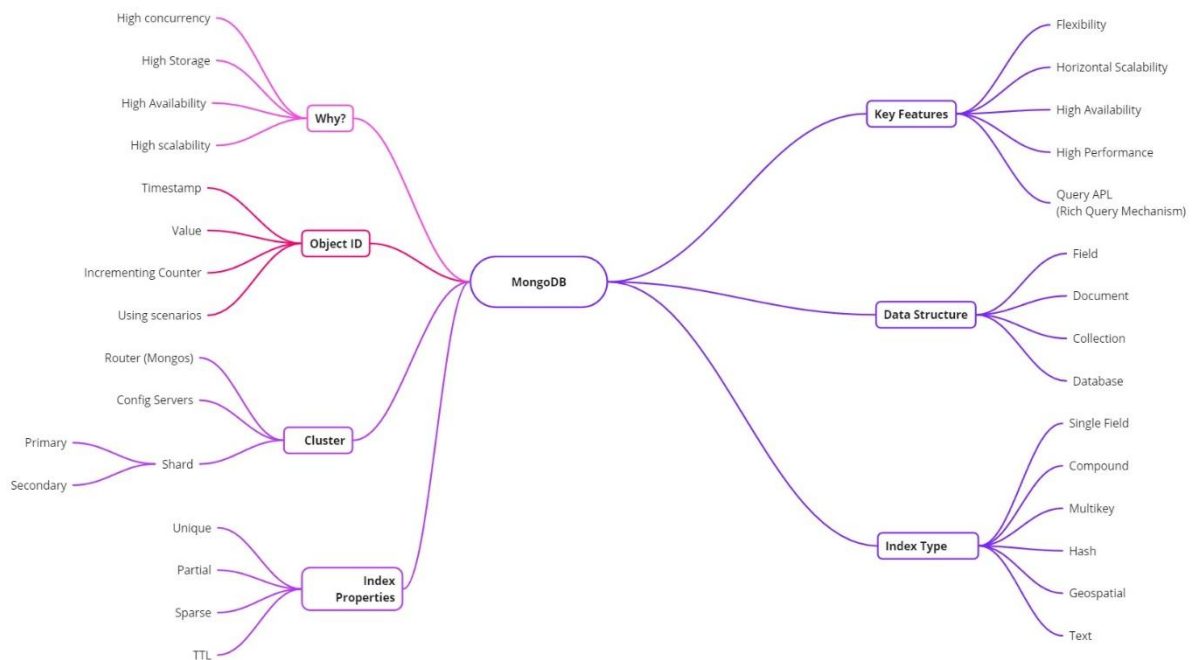
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Course: Big Data Analysis Technology

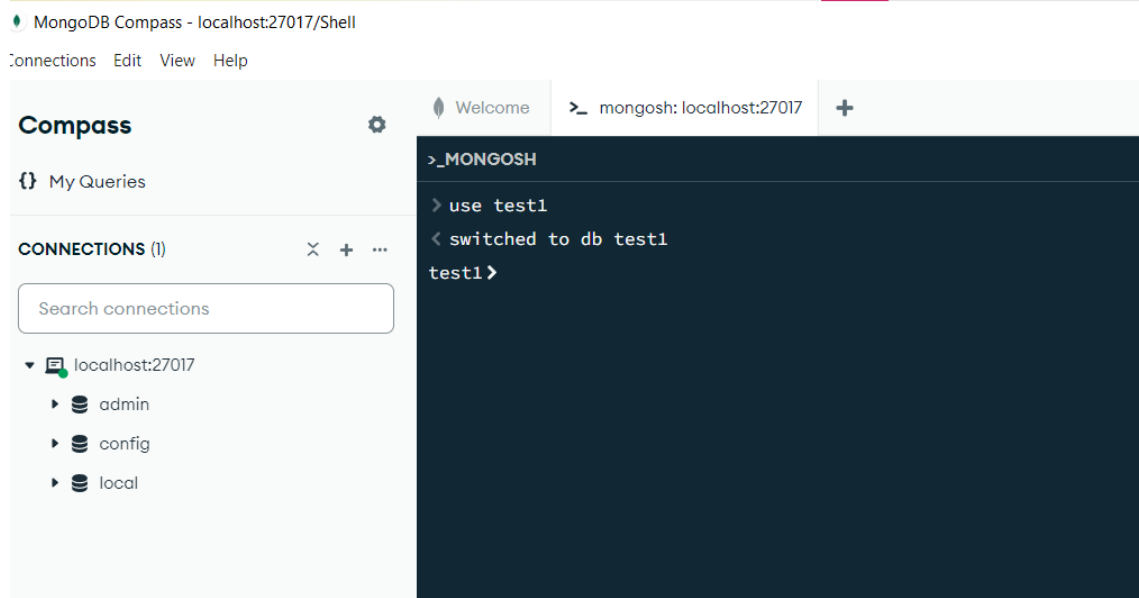
Experiment Topic: DocumentDB and MongoDB

Mind Map

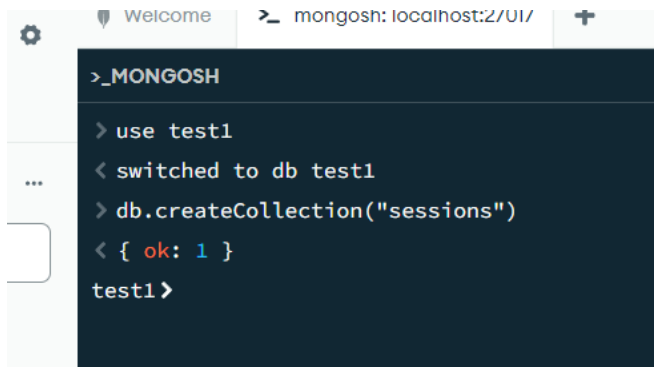


Experiment 1 (CRUD)

1. Starting MongoDB Shell and selecting a database “test1”



2. Creating a collection “sessions”



```
>_MONGOSH
> use test1
< switched to db test1
> db.createCollection("sessions")
< { ok: 1 }
test1>
```

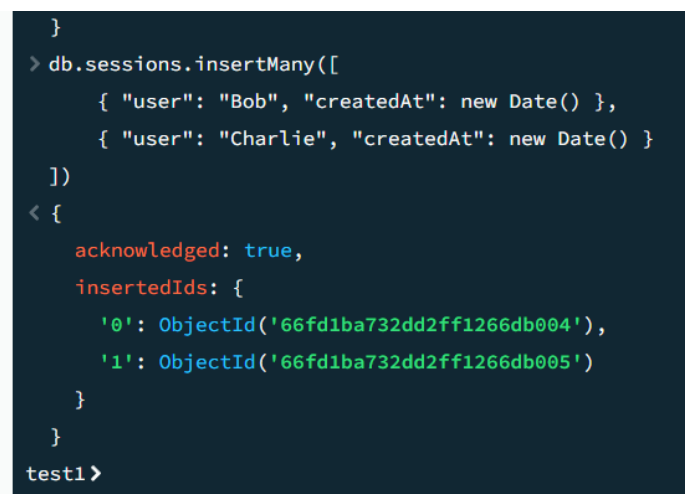
3. Creating Documents (Insert)



```
>_MONGOSH
> use test1
< switched to db test1
> db.createCollection("sessions")
< { ok: 1 }
> db.sessions.insertOne({ "user": "Alice", "createdAt": new Date() })
< {
  acknowledged: true,
  insertedId: ObjectId('66fd1b3432dd2ff1266db003')
}
test1>
```



```
}
> db.sessions.find().pretty()
< {
  _id: ObjectId('66fd1b3432dd2ff1266db003'),
  user: 'Alice',
  createdAt: 2024-10-02T10:06:44.248Z
}
test1>
```



```
}
> db.sessions.insertMany([
  { "user": "Bob", "createdAt": new Date() },
  { "user": "Charlie", "createdAt": new Date() }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('66fd1ba732dd2ff1266db004'),
    '1': ObjectId('66fd1ba732dd2ff1266db005')
  }
}
test1>
```

Using `db.sessions.find()` to find all the data in the database.

```

> db.sessions.find()
< {
  _id: ObjectId('66fd1b3432dd2ff1266db003'),
  user: 'Alice',
  createdAt: 2024-10-02T10:06:44.248Z
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db004'),
  user: 'Bob',
  createdAt: 2024-10-02T10:08:39.113Z
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db005'),
  user: 'Charlie',
  createdAt: 2024-10-02T10:08:39.113Z
}

```

4. Updating Documents (Update)

```

}
> db.sessions.updateOne(
  { "user": "Alice" }, // Query condition
  { $set: { "user": "Alice Smith" } } // Update content
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test1>

```

```

> db.sessions.find()
< {
  _id: ObjectId('66fd1b3432dd2ff1266db003'),
  user: 'Alice Smith',
  createdAt: 2024-10-02T10:06:44.248Z
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db004'),
  user: 'Bob',
  createdAt: 2024-10-02T10:08:39.113Z
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db005'),
  user: 'Charlie',
  createdAt: 2024-10-02T10:08:39.113Z
}

```

Now it's Alice Smith (not just Alice).

```

> db.sessions.updateMany(
  { "user": "Bob" },
  { $set: { "status": "active" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test1>

```

```

> db.sessions.find()
< {
  _id: ObjectId('66fd1b3432dd2ff1266db003'),
  user: 'Alice Smith',
  createdAt: 2024-10-02T10:06:44.248Z
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db004'),
  user: 'Bob',
  createdAt: 2024-10-02T10:08:39.113Z,
  status: 'active'
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db005'),
  user: 'Charlie',
  createdAt: 2024-10-02T10:08:39.113Z
}
test1>

```

Now Bob is active

5. Deleting Documents (Delete)

```

> db.sessions.deleteOne({ "user": "Alice Smith" })
< {
  acknowledged: true,
  deletedCount: 1
}
test1>

```

```
> db.sessions.find()
< {
  _id: ObjectId('66fd1ba732dd2ff1266db004'),
  user: 'Bob',
  createdAt: 2024-10-02T10:08:39.113Z,
  status: 'active'
}
{
  _id: ObjectId('66fd1ba732dd2ff1266db005'),
  user: 'Charlie',
  createdAt: 2024-10-02T10:08:39.113Z
}
test1>
```

```
> db.sessions.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 2
}
```

```
> db.sessions.find()
<
test1>
```

All the data has been deleted.

Experiment 2

1. Installing PyMongo

```
Terminal Local x + v
(venv) PS C:\Users\prett\PycharmProjects\rucharm> pip install pymongo
Collecting pymongo
  Obtaining dependency information for pymongo from https://files.pythonhosted.org/packages/cp311-win\_amd64.whl.metadata
    Downloading pymongo-4.10.1-cp311-cp311-win_amd64.whl.metadata (22 kB)
Collecting dnspython<3.0.0,>=1.16.0 (from pymongo)
  Obtaining dependency information for dnspython<3.0.0,>=1.16.0 from https://files.pythonhosted.org/packages/ython-2.6.1-py3-none-any.whl.metadata
    Downloading dnspython-2.6.1-py3-none-any.whl.metadata (5.8 kB)
```

2. Inserting the code on Python (with random seed 42)

```
1 import random
2 from datetime import datetime, timedelta
3 from faker import Faker
4 from pymongo import MongoClient
5
6 fake = Faker()
7 fake.seed_instance(42)
8 random.seed(42)
9 # MongoDB connection settings
10 mongo_client = MongoClient('mongodb://localhost:27017/')
11 db = mongo_client['test5'] # Replace with your database name
12 collection = db['test5'] # Replace with your collection name
13 collection.delete_many({})
14
```

```
Inserted 95000 data.
Inserted 96000 data.
Inserted 97000 data.
Inserted 98000 data.
Inserted 99000 data.
Successfully inserted 100000 data into MongoDB.
```

3. Aggregation

```

pipeline = [
    {
        "$group": {
            "_id": "$category",
            "count": {"$sum": 1}
        }
    }
]

result = list(collection.aggregate(pipeline))
for item in result:
    print(f"Category: {item['_id']}, Count: {item['count']}")

```

```

Category: Category B, Count: 25063
Category: Category C, Count: 25018
Category: Category D, Count: 24970
Category: Category A, Count: 24949

```

4. Calculating Average Price per Category

```

pipeline = [
    {
        "$group": {
            "_id": "$category",
            "average_price": {"$avg": "$price"}
        }
    }
]

result = list(collection.aggregate(pipeline))
for item in result:
    print(f"Category: {item['_id']}, Average Price: {item['average_price']:.2f}")

```

```

Category: Category D, Average Price: 254.45
Category: Category A, Average Price: 254.05
Category: Category B, Average Price: 254.22
Category: Category C, Average Price: 256.09

```

5. Counting Records by Tag

```
pipeline = [  
    {  
        "$unwind": "$tags"  
    },  
    {  
        "$group": {  
            "_id": "$tags",  
            "count": {"$sum": 1}  
        }  
    }  
]  
  
result = list(collection.aggregate(pipeline))  
for item in result:  
    print(f"Tag: {item['_id']}, Count: {item['count']}")
```

```
Tag: tag4, Count: 60175  
Tag: tag2, Count: 59666  
Tag: tag5, Count: 59969  
Tag: tag1, Count: 60126  
Tag: tag3, Count: 60189
```


Experiment 3

1. Measuring time without index (0.0456s).

```
#measure the time with no index
search_point = {
    "type": "Point",
    "coordinates": [-50.781527, -32.869214]
}
near_query = {
    "location": {
        "$geoWithin": {
            "$centerSphere": [search_point['coordinates'], 1000 / 6378.1]
        }
    }
}

import time
start_time = time.time()
cursor = collection.find(near_query)
for doc in cursor:
    print(doc)
end_time = time.time()
print(f"measure the time with no index: {end_time - start_time:.4f}s")
```

```
{ '_id': 'ab572fa7-9673-4675-97ad-657946908fc7', 'name': 'Heather Hall', 'description': 'North sense finally boy suggest back. Purpose know growth lot tree war.\nEven degree re
{ '_id': 'de3a0354-4a10-467d-884b-884271a5500b', 'name': 'Meghan Hansen', 'description': 'Area author need walk total military perform space. Miss military American run threat
{ '_id': '97b96d9c-2740-4e68-9512-99d97d30e5d3', 'name': 'Keith Yoder', 'description': 'Because feeling thing. Range manager strong begin subject price day.\nSpace tell purpose
{ '_id': '73765444-601a-4bb2-a40a-25f1d9b2bb13', 'name': 'Jeremiah Hudson', 'description': 'Set either rather strategy. They financial enter tree deal drop wind.\nReveal citize
{ '_id': 'aeeef89a5-5aad-4df1-b676-6c1289773ba7', 'name': 'Faith Vance', 'description': 'Your study car name part make allow determine. Loss throw heavy his determine instead ne
{ '_id': '10ca647d-eddf-46d9-aef8-a299ddb50a54', 'name': 'Samuel Sullivan', 'description': 'Thus him operation use a media community. Big what reflect fly impact learn.\nFeelin
{ '_id': '8968093a-4b5b-4027-a397-535c095b9885', 'name': 'Brittany Casey', 'description': 'Discuss specific instead. Hear agency arrive pattern us. Notice religious wish list d
{ '_id': '0583bf66-a1f0-4d40-babe-e6fc483eaf50', 'name': 'Mary Griffin', 'description': 'Happy mission ever statement agree country. None class out wide. Rule production agent
measure the time with no index: 0.0456s
```

2. Creating an index and calculating time with it (0.0370s).

```
# create index of 2d location
collection.create_index([('location', '2dsphere')])
```

```
start_time = time.time()
cursor = collection.find(near_query)
for doc in cursor:
    print(doc)
end_time = time.time()
print(f"measure the time after creating index: {end_time - start_time:.4f}s")
```

```
{ '_id': 'ab572fa7-9673-4675-97ad-657946908fc7', 'name': 'Heather Hall', 'description': 'North sense finally boy suggest back. Purpose know growth
{ '_id': 'de3a0354-4a10-467d-884b-884271a5500b', 'name': 'Meghan Hansen', 'description': 'Area author need walk total military perform space. Mis
{ '_id': '97b96d9c-2740-4e68-9512-99d97d30e5d3', 'name': 'Keith Yoder', 'description': 'Because feeling thing. Range manager strong begin subject
{ '_id': '73765444-601a-4bb2-a40a-25f1d9b2bb13', 'name': 'Jeremiah Hudson', 'description': 'Set either rather strategy. They financial enter tree
{ '_id': 'aeeef89a5-5aad-4df1-b676-6c1289773ba7', 'name': 'Faith Vance', 'description': 'Your study car name part make allow determine. Loss throw
{ '_id': '10ca647d-eddf-46d9-aef8-a299ddb50a54', 'name': 'Samuel Sullivan', 'description': 'Thus him operation use a media community. Big what re
{ '_id': '8968093a-4b5b-4027-a397-535c095b9885', 'name': 'Brittany Casey', 'description': 'Discuss specific instead. Hear agency arrive pattern u
{ '_id': '0583bf66-a1f0-4d40-babe-e6fc483eaf50', 'name': 'Mary Griffin', 'description': 'Happy mission ever statement agree country. None class o
measure the time after creating index: 0.0370s
```

Experiment 4

1. Creating an index of text and measuring the time after creation (0.1230s)

```
# create index of text
collection.create_index([('description', 'text')])

# measure the time after creating index
import time
start_time = time.time()
search_query = {'$text': {'$search': 'administration'}}
cursor = collection.find(search_query)
for doc in cursor:
    print(doc)
end_time = time.time()
print(f"measure the time after creating index: {end_time - start_time:.4f}s")
num_docs = collection.count_documents(search_query)
print(f"Number of documents found:{num_docs}")
```

```
{ '_id': '76f18465-05a7-4bb9-892f-b3e044ac2d95', 'name': 'Jonathan Frazier', 'description': 'Put lay program southern produce. Front seek hear yes wide pul' }
{ '_id': '84f15969-3c1b-4b2f-8384-fdf369300821', 'name': 'Michael Forbes', 'description': 'Site offer room hard charge condition country TV. Single join br' }
{ '_id': '56cb6d21-a380-4633-acc1-6b52b3797911', 'name': 'Christopher Lynn', 'description': 'Hotel low goal doctor. Feel ahead inside budget group. Candida' }
{ '_id': 'c7907aec-e61b-4d4f-9994-af65bac289e2', 'name': 'Gabrielle Thompson', 'description': 'Part determine concern player road cup. Speech hair perform' }
{ '_id': '08d8d6df-6d6b-470f-8db1-505e4b669a1d', 'name': 'David Reyes', 'description': 'Stand stand outside stuff moment. Prevent wind case say boy politic' }
{ '_id': '23d24b8f-1cae-42f7-8b84-a168d3565b92', 'name': 'Jessica Eaton', 'description': 'President now discover war film kitchen radio see. West build tot' }
{ '_id': '2d2adb9d-9e1f-4c9f-b16d-7f80427d192b', 'name': 'Keith Garcia', 'description': 'Alone bag a coach minute. Argue drug theory cover move may. Examl' }
{ '_id': 'c74e3be9-f928-4eaa-aeb7-fe4a9b318314', 'name': 'Olivia Maynard', 'description': 'Black specific may rock. Benefit number want home end. Two rock' }
measure the time after creating index: 0.1230s
Number of documents found:2167
```

2. Finding the current index now

```
print("current Index: ")
for index_info in collection.list_indexes():
    print(index_info)
```

```
current Index:
SON([('v', 2), ('key', SON([('id', 1]))), ('name', '_id-')])
SON([('v', 2), ('key', SON([('location', '2dsphere')]))), ('name', 'location_2dsphere'), ('2dsphereIndexVersion', 3)])
SON([('v', 2), ('key', SON([('fts', 'text'), ('_ftsx', 1)])), ('name', 'description_text'), ('weights', SON([('description', 1)])), ('default_language', 'english'), ('language_override', 'language'), ('textIndexVersion', 3)])
```